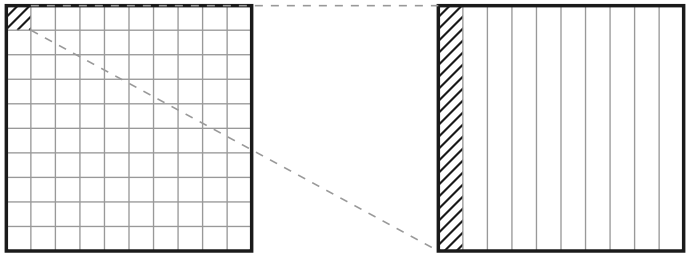
Thousandths

Cut a hundredth into 10 equal parts. Each part is a thousandth.



1 hundredth = 0.01 1 thousandth = 0.001

10 thousandths = 1 hundredth
100 thousandths = 1 tenth
1,000 thousandths = 1 whole

2.457 in the place-value chart

thousands	hundreds	tens	ones		tenths	hundredths	thousandths
			2	.	4	5	7

The 7 is in the thousandths place, so it is worth 0.007.

$$2 + 0.4 + 0.05 + 0.007 = 2.457$$

Say it

2.457 = two and four hundred fifty-seven thousandths

Say the whole number, then "and" for the point, then the digits after it and the last place: 0.06 = six hundredths, 0.045 = forty-five thousandths.

Your turn!

in 5.382 the 8 is worth **0.08**

For teachers

Goal: read and write decimals to thousandths in the chart, in expanded form and in words (5.NBT.3a). Say "A thousandth is one hundredth cut into 10." Read the point as "and", never "point".

Thousandths: Practice

Each place is worth 10 of the place to its right.

Part 1 Read the chart Write the number.

1)

th	h	t	o		tth	hth	thth
			3	.	4	6	2

the number is **3.462**

2)

th	h	t	o		tth	hth	thth
			0	.	5	0	8

the number is **0.508**

3)

th	h	t	o		tth	hth	thth
		1	4	.	0	7	

the number is **14.07**

4)

th	h	t	o		tth	hth	thth
			6	.	0	3	9

the number is **6.039**

5)

th	h	t	o		tth	hth	thth
			0	.	7	1	5

the number is **0.715**

Part 2 Fill the chart Write each digit in its place.

6) **2.184**

th	h	t	o		tth	hth	thth
			2	.	1	8	4

7) **0.096**

th	h	t	o		tth	hth	thth
			0	.	0	9	6

8) **31.5**

th	h	t	o		tth	hth	thth
		3	1	.	5		

Thousandths: Practice (continued)

Part 2 Fill the chart (continued)

9) **7.603**

th	h	t	o		tth	hth	thth
			7	.	6	0	3

10) **0.042**

th	h	t	o		tth	hth	thth
			0	.	0	4	2

Part 3 What is the digit worth? Some help: look at the place after the point.

11) in 4.276 the 6 is worth **0.006**

12) in 0.835 the 8 is worth **0.8**

13) in 9.162 the 6 is worth **0.06**

14) in 1.409 the 9 is worth **0.009**

15) in 6.358 the 5 is worth **0.05**

16) in 0.274 the 2 is worth **0.2**

17) in 2.917 the 7 is worth **0.007**

18) in 5.643 the 4 is worth **0.04**

19) in 8.521 the 5 is worth **0.5**

20) in 3.084 the 8 is worth **0.08**

Thousandths: Practice (continued)

Part 3 What is the digit worth? (continued)

21) in 7.395 the 5 is worth **0.005**

22) in 0.769 the 7 is worth **0.7**

Part 4 Put it together No help this time.

23) 5 + 0.3 + 0.02 + 0.006 = **5.326**

24) 0.4 + 0.07 + 0.001 = **0.471**

25) 2 + 0.08 = **2.08**

26) 8 + 0.9 + 0.004 = **8.904**

27) 13 + 0.2 + 0.05 + 0.007 = **13.257**

28) 0.06 + 0.003 = **0.063**

29) 4 + 0.6 + 0.01 = **4.61**

30) 9 + 0.03 + 0.005 = **9.035**

31) 6 + 0.7 + 0.002 = **6.702**

32) 1 + 0.5 + 0.09 + 0.008 = **1.598**

Thousandths: Practice (continued)

Part 5 Take it apart No help this time. Write each part.

33) 3 + 0.5 + 0.01 + 0.007 = 3.517

34) 0.9 + 0.02 + 0.008 = 0.928

35) 6 + 0.4 + 0.003 = 6.403

36) 7 + 0.06 = 7.06

37) 2 + 0.3 + 0.09 + 0.001 = 2.391

38) 0.8 + 0.005 = 0.805

Part 6 Fractions to decimals No help this time.

39) $\frac{4}{10} + \frac{8}{100} + \frac{3}{1{,}000} = 0.483$

40) $2 + \frac{7}{10} + \frac{1}{100} + \frac{6}{1{,}000} = 2.716$

41) $\frac{5}{100} + \frac{9}{1{,}000} = 0.059$

42) $5 + \frac{3}{10} + \frac{2}{1{,}000} = 5.302$

43) $\frac{6}{10} + \frac{4}{100} = 0.64$

44) $1 + \frac{9}{10} + \frac{7}{1{,}000} = 1.907$

Thousandths: Practice (continued)

Part 6 Fractions to decimals (continued)

45) $\frac{2}{100} + \frac{5}{1{,}000} =$ **0.025**

46) $4 + \frac{1}{10} + \frac{3}{100} + \frac{8}{1{,}000} =$ **4.138**

Part 7 Words to numbers No help this time.

47) **two and forty-five thousandths =** **2.045**

48) **three hundred eighteen thousandths =** **0.318**

49) **seven and six tenths =** **7.6**

50) **nine thousandths =** **0.009**

51) **twelve and fifty-three hundredths =** **12.53**

52) **four and two hundred seven thousandths =** **4.207**

53) **eighty-six hundredths =** **0.86**

54) **three and twelve thousandths =** **3.012**

Thousandths: Practice (continued)

Part 8 How many thousandths? No help this time.

55) 0.4 = 400 thousandths

56) 0.27 = 270 thousandths

57) 0.6 = 600 thousandths

58) 0.051 = 51 thousandths

59) 0.93 = 930 thousandths

60) 0.2 = 200 thousandths

61) 0.307 = 307 thousandths

62) 0.08 = 80 thousandths

63) 0.75 = 750 thousandths

Part 9 What is the digit worth?

64) in 1.789 the 9 is worth 0.009

65) in 0.363 the 3 is worth 0.003

Thousandths: Practice (continued)

Part 9 What is the digit worth? (continued)

66) in 6.752 the 7 is worth **0.7**

67) in 4.552 the 5 is worth **0.5**

68) in 0.346 the 4 is worth **0.04**

69) in 1.833 the 3 is worth **0.003**

70) in 7.233 the 2 is worth **0.2**

71) in 5.125 the 1 is worth **0.1**

72) in 1.635 the 5 is worth **0.005**

73) in 8.768 the 8 is worth **0.008**

74) in 2.734 the 3 is worth **0.03**

75) in 6.218 the 2 is worth **0.2**

76) in 3.461 the 1 is worth **0.001**

77) in 0.302 the 3 is worth **0.3**

Thousandths: Practice (continued)

Part 9 What is the digit worth? (continued)

78) in 4.164 the 1 is worth **0.1**

79) in 9.909 the 9 is worth **0.9**

80) in 4.663 the 6 is worth **0.06**

81) in 9.184 the 1 is worth **0.1**

82) in 4.199 the 1 is worth **0.1**

83) in 3.988 the 9 is worth **0.9**

84) in 9.701 the 7 is worth **0.7**

85) in 2.191 the 1 is worth **0.1**

86) in 7.999 the 9 is worth **0.009**

87) in 1.185 the 5 is worth **0.005**

Thousandths: Practice (continued)

Part 10 Put it together

88) $3 + 0.5 + 0.01 + 0.009 =$ 3.519

89) $8 + 0.6 + 0.002 =$ 8.602

90) $12 + 0.8 + 0.01 + 0.006 =$ 12.816

91) $4 + 0.3 + 0.05 + 0.005 =$ 4.355

92) $12 + 0.2 + 0.01 + 0.004 =$ 12.214

93) $6 + 0.6 + 0.002 =$ 6.602

94) $13 + 0.7 + 0.03 + 0.001 =$ 13.731

95) $10 + 0.9 + 0.01 + 0.001 =$ 10.911

96) $1 + 0.7 + 0.01 + 0.002 =$ 1.712

97) $10 + 0.4 + 0.05 + 0.005 =$ 10.455

98) $12 + 0.3 + 0.03 + 0.006 =$ 12.336

99) $3 + 0.3 + 0.01 + 0.005 =$ 3.315

Thousandths: Practice (continued)

Part 10 Put it together (continued)

100) $1 + 0.9 + 0.07 + 0.005 =$ **1.975**

101) $11 + 0.4 + 0.03 + 0.003 =$ **11.433**

102) $10 + 0.1 + 0.007 =$ **10.107**

103) $14 + 0.4 + 0.05 + 0.009 =$ **14.459**

104) $15 + 0.7 + 0.09 + 0.003 =$ **15.793**

105) $14 + 0.08 + 0.003 =$ **14.083**

106) $9 + 0.2 + 0.04 + 0.007 =$ **9.247**

107) $11 + 0.9 + 0.07 + 0.004 =$ **11.974**

108) $12 + 0.3 + 0.02 + 0.005 =$ **12.325**

109) $5 + 0.1 + 0.02 + 0.002 =$ **5.122**

110) $1 + 0.4 + 0.06 + 0.007 =$ **1.467**

111) $12 + 0.3 + 0.07 + 0.008 =$ **12.378**

Thousandths: Practice (continued)

Part 11 Remember? Problems from earlier packets.

112) $6.3 \times 100 =$ **630**

113) $740 \div 100 =$ **7.4**

114) $10 \times 10 \times 10 =$ **1,000**

115) 0.48 **<** 0.5

116) $7 \times \frac{2}{9} = \frac{14}{9}$

117) $47 \times 23 =$ **1,081**

Part 12 Word problems Think about the places.

118) A runner's time is 12.345 seconds. What is the 4 worth?

Number sentence **the 4 is in the hundredths place**

Answer **0.04** seconds

Draw or show your work

119) A bag of rice weighs 2 kg and 75 thousandths of a kg. Write it as a decimal.

Number sentence **2 + 0.075 = 2.075**

Answer **2.075** kg

Draw or show your work

120) A nail is 0.045 m long. How many thousandths of a meter is that?

Number sentence **0.045 = 45 thousandths**

Answer **45** thousandths

Draw or show your work

Thousandths: Practice (continued)

Part 12 Word problems (continued)

121) A seed weighs 0.008 g. How much do 10 of these seeds weigh?

Number sentence $0.008 \times 10 = 0.08$

Answer **0.08** g

Draw or show your work

122) Challenge: A number has 4 ones, 7 thousandths, 2 hundredths and no tenths. What is the number?

Number sentence $4 + 0.02 + 0.007 = 4.027$

Answer **4.027**

Draw or show your work

Check-In: Thousandths

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Fill the chart

1) **3.519**

th	h	t	o		tth	hth	thth
			3	.	5	1	9

Part 2 What is the digit worth?

2) in 7.263 the 2 is worth **0.2**

3) in 0.724 the 4 is worth **0.004**

Part 3 Put it together

4) 4 + 0.5 + 0.01 + 0.007 = **4.517**

5) 3 + 0.2 + 0.04 + 0.009 = **3.249**

Part 4 Words to numbers

6) six thousandths = **0.006**

7) five and three hundred two thousandths = **5.302**

Part 5 How many thousandths?

8) 0.605 = **605** thousandths

9) 0.5 = **500** thousandths